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ABSTRACT

For the design and certification of wind turbines, it is essential to provide fast and accurate unsteady aerodynamic load prediction models for the whole operational range of angle of attack, up to 180° for vertical-axis and 90° for horizontal-axis wind turbines. This work describes a computationally efficient unsteady forces prediction model based on a deep learning approach, namely the bidirectional long short-term memory (BiLSTM) algorithm, for an airfoil pitched over the full operational range of angles of attack up to 180°. No model has been developed to capture the unsteady forces at high angles of attack. Novel features based on operating conditions and the steady polars of the airfoil are used as inputs for the BiLSTM model. Direct measurements of steady and unsteady forces on a NACA 0021 airfoil model were conducted at reduced frequencies up to 0.075 and a Reynolds number of 120 000 in an open-jet wind tunnel for model learning and testing. The unsteady forces vary significantly from the steady values at high pitching amplitudes and post-stall angles, which, if not accounted for when simulating wind turbine performance, would result in inaccurate predictions. Furthermore, measurements revealed the effect of unsteady vorticity development and shedding on aerodynamic forces under forward and reverse flow conditions. The BiLSTM model is capable of capturing the underlying physics of unsteady aerodynamic forces under extreme operating conditions.

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I. INTRODUCTION

Wind turbines encounter unsteady conditions in the form of flow separation, dynamic stall, flow reattachment, and unsteady attached flow during most of their lifespan, resulting in substantial nonlinearity and hysteresis in the aerodynamic forces. Flow unsteadiness is the consequence of a variety of causes, such as tower shadow effects, yaw misalignment and wind shear, atmospheric turbulence, and the cyclic nature of vertical-axis wind turbine (VAWT) operation.[1,2] These result in unsteady variations in the effective angles of attack, leading to nonlinear, time-dependent aerodynamic forces and fatigue loads on turbine blades. It is common to quantify the flow unsteadiness for an oscillating airfoil by the reduced frequency, $k = \omega c/(2V_\infty)$, where ω is the pitching frequency, c is the airfoil chord, and V_∞ is the wind speed.[3] The angles of attack can reach 90° for horizontal-axis wind turbines (HAWTs) and up to 180° for VAWTs.[4] Nevertheless, the literature lacks unsteady airfoil data and prediction models over the whole operating range of wind turbine blades.[5] Experimental unsteady airfoil data and high-fidelity models for high angles of attack are needed to develop accurate unsteady load predictions that can extend the life of wind turbines and improve the economics of harvesting wind energy.

Unsteady aerodynamic forces can be modeled using computational fluid dynamics (CFD), empirical modeling, and data-driven models. CFD relies on physics modeling and requires significant computational resources, which hinders its utilization in iterative design or load-certification. Furthermore, the computed unsteady aerodynamic forces are often in disagreement with experimental data during flow separation and reattachment.[6,7] Empirical models are the common approach adopted in academia and industry to reproduce dynamic stall loads for wind turbine design and aeroelastic simulations, in which unsteady aerodynamic forces are modeled based on physical intuition and engineering assumptions. However, no such empirical model has been developed to predict unsteady aerodynamic loads on airfoils or wind turbine blades for the entire operating range of angles

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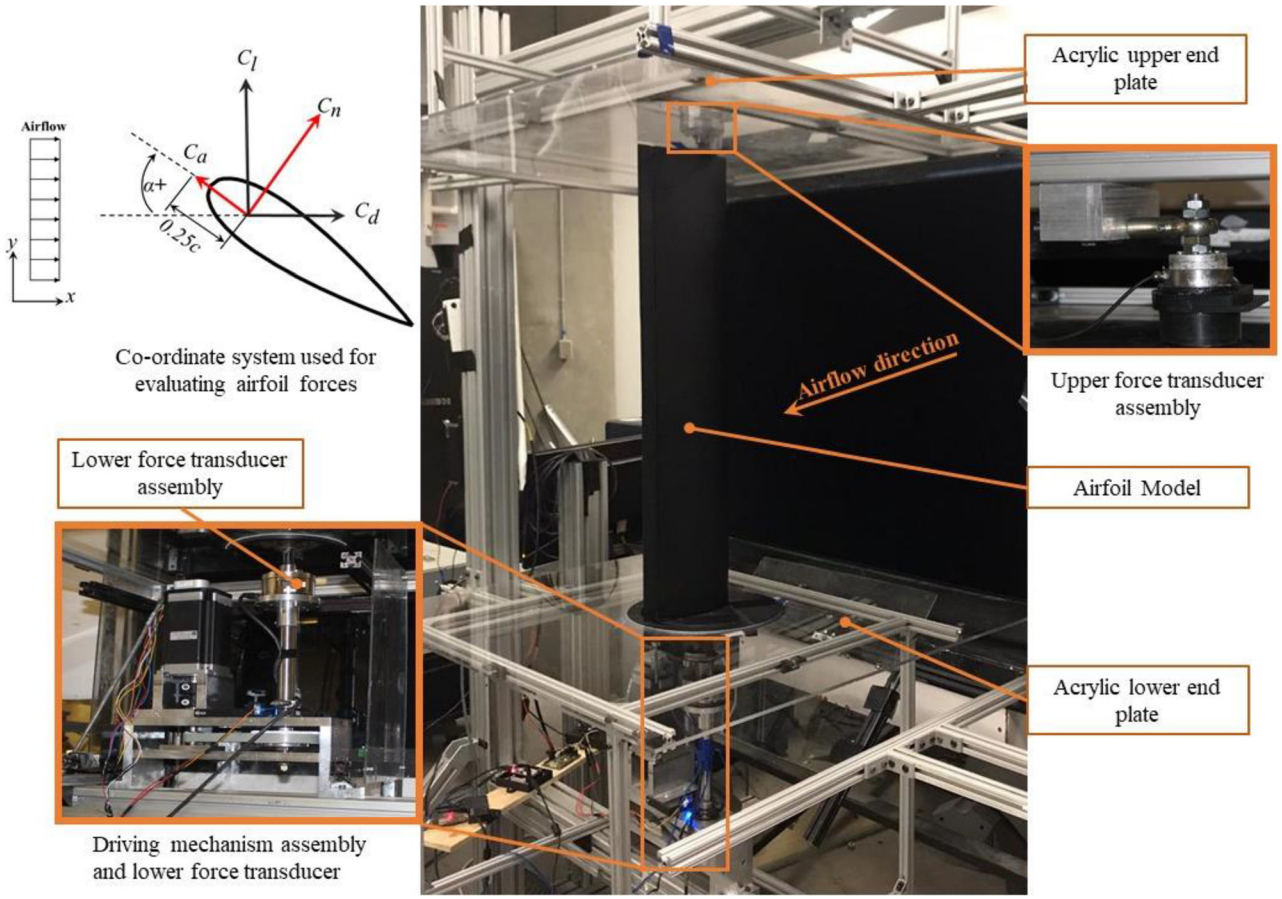


FIG. 1. The wind tunnel experimental setup and co-ordinate system used for evaluating airfoil forces. The top and bottom acrylic plates were aligned to the exit of the wind tunnel contraction. The contraction outlet was $1 \times 1\ m^2$, and the airfoil quarter-chord was 0.45 m from the contraction exit.

of attack from 0° to 180°. For instance, OpenFAST, a wind turbine analysis code developed by the U.S. National Renewable Energy Laboratory, deactivates the unsteady forces computed by the unsteady aerodynamic empirical models when the estimated effective angle of attack exceeds 45°.[8] A modified version of the ONERA (Office National d'études et de Recherches Aérospatiales) dynamic stall model for several airfoils was recently developed by Mohamed and Wood,[2] with the goal of modeling the load variations associated with the shedding of multiple dynamic stall vortices (DSVs) up to angles of attack of 30° only. DSVs are the main unsteady flow feature associated with dynamic stall. They form and detach from near the leading edge as the load changes, and their passage over the airfoil further influences the lift and drag. At elevated angles of attack, it is difficult to tune empirical models over wide operating conditions, as the flow response is highly nonlinear.[9] It was the difficulty in tuning the model of Ref. 2 for the present data that motivated us to investigate machine learning alternatives.

Data-driven models have been demonstrated to be highly effective in a broad range of applications, including regression and classification problems, in which it is difficult to find the underlying physics.[10–14] Recent advances in data science and machine learning have opened up new avenues for addressing otherwise intractable aerodynamics and fluid mechanics challenges. Applications in aerodynamics include airfoil shape design and optimization,[15,16] flow estimation and reconstruction,[17,18] turbulence modeling for the Reynolds averaged Navier–Stokes equations,[19] aeroelasticity,[20] and load estimation for smart airfoils.[21,22]

To overcome unsteady aerodynamic modeling challenges, many machine learning techniques have been developed, including support vector machine (SVM),[23] neural network (NN),[24,25] recurrent neural network (RNN) with gated recurrent unit (GRU),[26] long short-term memory (LSTM),[27,28] and adaptive network-based Fuzzy inference system (ANFIS).[29] Machine learning algorithms may be trained using both force measurements in wind tunnels[1,9] and from CFD simulations.[24,26,30] All these models were trained to predict unsteady forces on airfoil models at angles of attack lower than 45°. Unsteady aerodynamic forces are time series data that exhibit an aerodynamic lag in the flow separation and reattachment on a pitched airfoil, causing hysteresis in the unsteady forces.[2] An accurate deep learning algorithm must reproduce the time histories of the circulatory and noncirculatory components generated by variations in effective angles of attack caused by varying inflow conditions and blade structural response.[31] The time dependencies in the unsteady forces can be learnt by a RNN

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algorithm without increasing model complexity due to weight sharing.[32] Because the RNN algorithm is trained through backpropagation through time, gradients have a tendency to vanish or explode, which deteriorates the algorithm performance.[33] Consequently, the RNN algorithm may not capture long-term dependencies. To tackle this problem, the LSTM algorithm, a variant of RNN, has been developed by including an explicit memory cell that utilizes nonlinear gate units to regulate the flow of information and learn the long-term dependencies.[34,35] As a unique variant of the LSTM algorithm, the bidirectional long short-term memory or BiLSTM algorithm has been introduced to increase the learning capability of the long-term dependencies in time-history data.[36] The fundamental component of BiLSTM is the LSTM, which consists of forward-direction LSTM and backward-direction LSTM, so it can track both the past and future status for every sequence sample by internally modeling the global dependencies using recurrent connections.[34]

Representative features can be constructed in terms of the operating conditions and the airfoil geometry. Operating conditions include the angle of attack (α), the pitching rate ($\dot{\alpha} = d\alpha/dt$), and the pitching acceleration ($\ddot{\alpha} = d^2\alpha/dt^2$).[9] The airfoil geometry can be parameterized with a universal parametric geometric representation or geometry variables, such as model thickness and camber line co-ordinates.[9,27] However, these geometric representation techniques lack critical information about the steady load curves, such as the static stall angle (α_{ss}) and load hysteresis that have strong relationships to the dynamic stall forces. Not representing these characteristics in the constructed features set requires the machine learning algorithm to differentiate between attached and separated flow regimes and to generate the unsteady force hysteresis. Extending the application of the machine learning framework to include fluid dynamics at high angles of attack exacerbates the problem's complexity and makes it extremely nonlinear. This necessitates the extraction of novel discriminative features to capture the wide range of operating conditions and airfoil performance in a way that is not achievable with either conventional unsteady modeling approaches or previous data-driven models.

The purpose of this work is to provide one of the first applications of deep learning algorithms, specifically the BiLSTM algorithm, for predicting the unsteady aerodynamic forces on an oscillating airfoil model for the entire operating range. The BiLSTM was chosen because, in comparison to the LSTM, it offers extra training by traveling both forward and backward through time-history data.[37] The proposed BiLSTM algorithm is trained using a subset of experimental data measured in a wind tunnel for a NACA 0021 airfoil model and then validated against the remaining data set. We believe that the present experiments provide a significant challenge for machine learning given the complex aerodynamic processes involved with the large

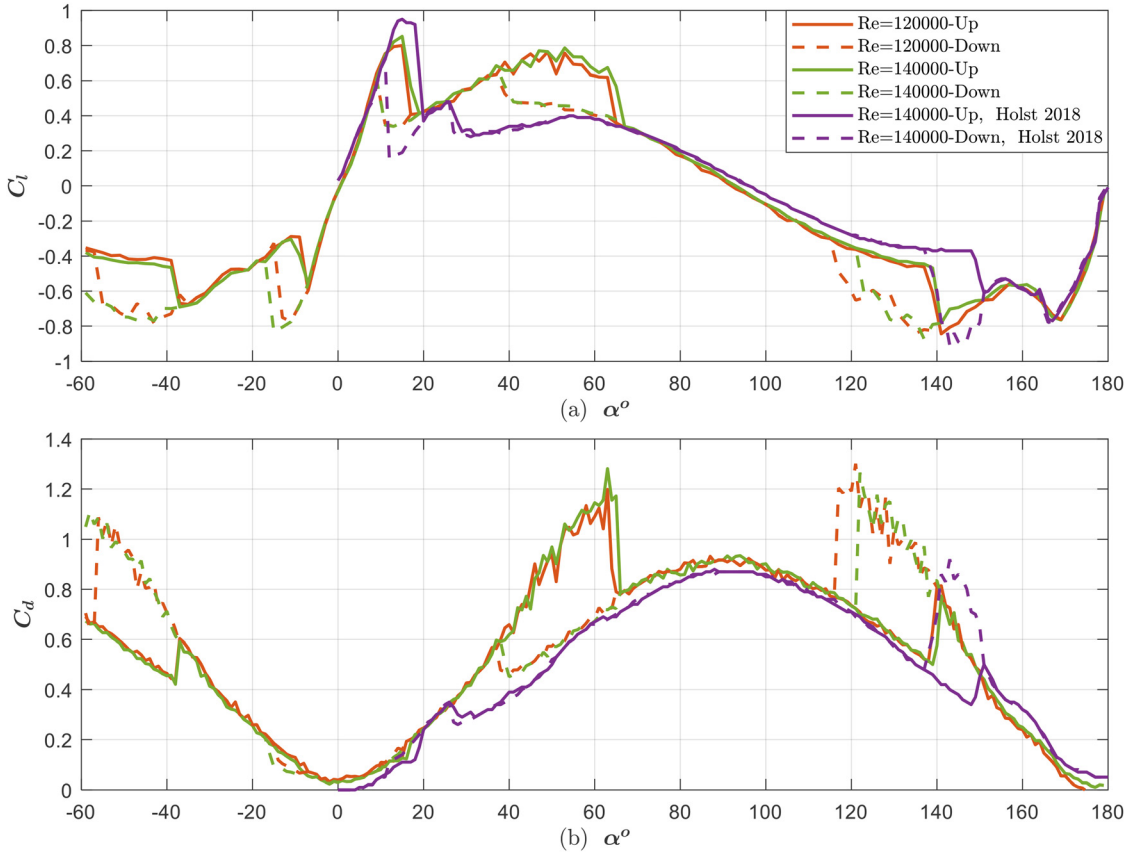


FIG. 2. Validation of the steady measurements using work of Holst *et al.*[4]: (a) C_l and (b) C_d.

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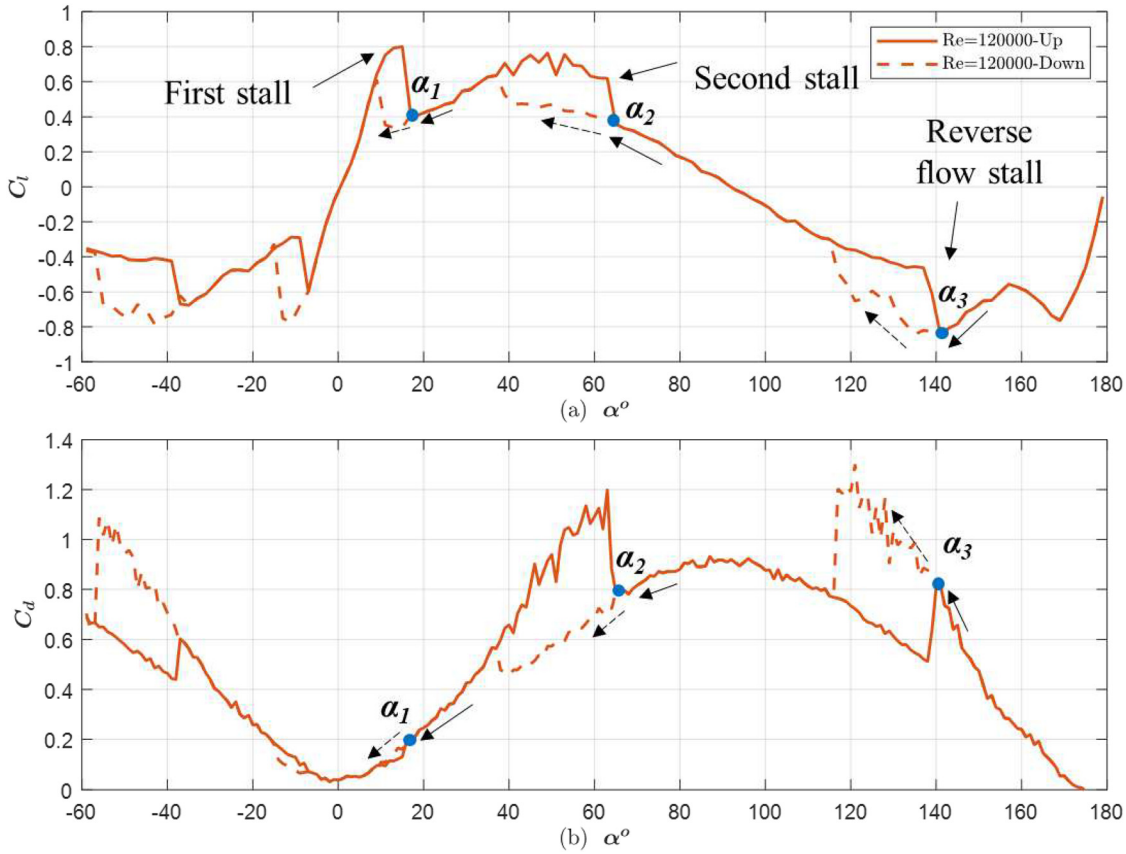


FIG. 3. Conditions for defining the steady curves as input for the BiLSTM dynamic stall model: (a) C_l and (b) C_d. The subscript indicates the number of the static stall.

range of angles and frequencies. Our approach is to make the model as general as possible; in other words, one model is used for all cases, and it will be shown that this approach works well in general. There are some features at high angles that the model does not accurately reproduce, but these cases are limited.

This work is structured as follows: Sec. II describes the experimental setup utilized to actuate the airfoil model and measure the unsteady aerodynamic forces directly in the wind tunnel. Section III outlines the key physics of the airfoil aerodynamic behavior at high angles of attack. The details of the proposed BiLSTM model, including the designed set of features, are discussed in Sec. IV, while the results are presented in Sec. V. In Sec. VI, the conclusions and suggestions for future research are outlined.

II. EXPERIMENT OVERVIEW

The wind tunnel experiment was carried out in a $1 \times 1\,m^2$ wind tunnel at the University of Calgary. A NACA 0021 airfoil model was used, with a chord length of $c = 0.185$ m, an aspect ratio of 4.5 and a blockage ratio of 18.5% at $\alpha = 90°$. The airfoil model consisted of five 3D-printed sections glued together on a 16 mm steel shaft. The final assembly was sanded and coated with epoxy for a smooth surface finish. The experiment was conducted at $V_\infty = 10$ m/s, corresponding to Reynolds number, $Re = 120\,000$, with a turbulence intensity level of 0.3%. A timing-belt system, driven by a stepper motor, was used to pitch the airfoil model about the quarter-chord location, as illustrated in Fig. 1. The stepper motor was equipped with an optical encoder to measure α with a resolution of 0.0625°. The model was mounted vertically at the center of the wind tunnel test section, between two ATI (Industrial Automation) multi-axis force sensors, as illustrated in Fig. 1. The capacity of the lower force sensor was 32 N in the x- and y-directions, with a force resolution of 0.006 25 N, while the upper force transducer had a capacity of 20 N in the x- and y-directions, with a resolution of 0.005 N. The upper force transducer was mounted on a spherical rod end to rotate freely with the airfoil model. A National Instruments data acquisition card with a 16-bit resolution, NI-DAQ 6212, was connected to each force sensor. The two cards were synchronized to commence acquisition at the same time using an external trigger. Two horizontals $0.9 \times 0.9\,m^2$ acrylic end plates were used to alleviate the free-end effects. The end plates were equipped with a 3D-printed superelliptic leading edge to reduce the probability of flow separation at the leading edge.[38] No vertical end plates were used. The force transducers measured the total loads acting on the airfoils, including the inertial, frictional, and aerodynamic forces. For every test run, the true aerodynamic responses were found by subtracting the force measurements when the wind tunnel was on and off. The normal force, C_n, and chord-axis force, C_a, were measured and

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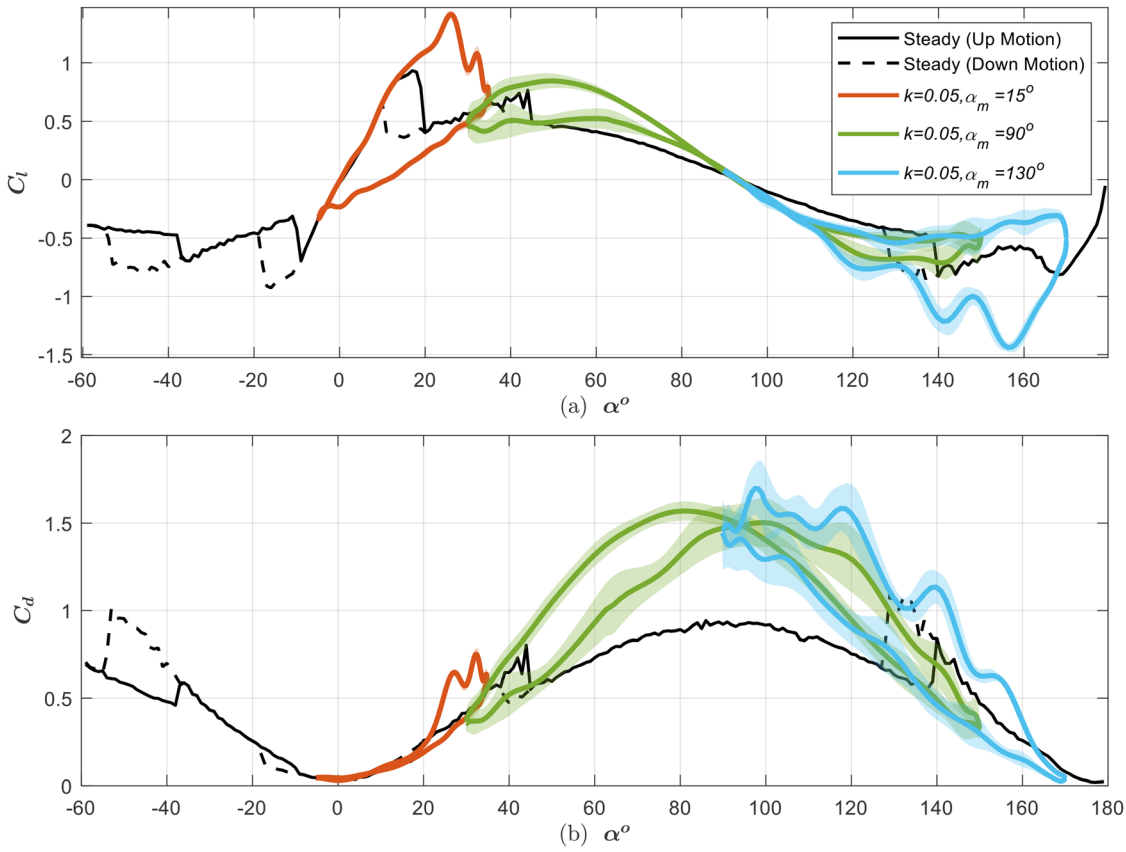


FIG. 4. Impact of dynamic stall vortex development and shedding on aerodynamic unsteady loads at mean angle of attack of $\alpha_m = 15°$ (red lines), 90° (green), and 130° (blue) and pitching amplitude of $\alpha_a = 60°$: (a) C_l and (b) C_d. The shading indicates the standard deviation over the 20 measurement cycles.

computed by summing the force readings from the two force transducers according to

$$C_n = \frac{F_{n1}+F_{n2}}{0.5\rho V_\infty^2 c} \tag{1}$$

and

$$C_a = \frac{F_{a1}+F_{a2}}{0.5\rho V_\infty^2 c}, \tag{2}$$

where the subscripts 1 and 2 represent the lower and the upper force transducer readings. Subsequently, lift and drag were computed as

$$C_l = C_n \cos\alpha + C_a \sin\alpha \tag{3}$$

and

$$C_d = C_n \sin\alpha - C_a \cos\alpha. \tag{4}$$

Because the flow was not constrained in the direction of the lift, and no effective blockage corrections for dynamic testing exist, the results are presented without adjustment. Furthermore, dynamic stall measurements are minimally affected by blockage effects and are qualitatively comparable after any correction.[39,40] Finally, the primary goal is to explore the application of deep learning algorithms for predicting unsteady forces at high angles of attack. The airfoil was pitched sinusoidally according to $\alpha = \alpha_m + \alpha_a \sin(k\tau - \pi/2)$ at four reduced frequencies, namely $k = 0.0125$, 0.025, 0.05, and 0.075, where the nondimensional time τ is computed as $\tau = 2V_\infty t/c$ and time t in seconds. The measurements were conducted at mean angles of attack of $\alpha_m = 0°$, 15°, 30°, 50°, 90°, and 130°, with a pitching amplitude range of $\alpha_a = 10°$–60°, to cover the full rotation of the airfoil and to investigate unsteady aerodynamic loads around load hysteresis regimes of the steady polars. The unsteady aerodynamic forces were phase averaged over a total of 20 cycles and presented along with the standard deviation indicated by shading on the subsequent figures.

III. STEADY AND UNSTEADY AERODYNAMIC LOADING

Steady measurements were made prior to the unsteady measurements. It is rare to find experimental data acquired under the same experimental conditions as the presented work for the NACA 0021 airfoil model pitched over the entire operating range up to 180°. Therefore, steady measurements were made for comparison with the data of Holst *et al.*[4] at $Re = 140\,000$, as shown in Fig. 2. It is noted that[4] determined the lift and drag by measuring only the airfoil surface pressure. This is likely to under-estimate the drag especially at low absolute angles of attack which led to discrepancies with the present data. The present steady and unsteady

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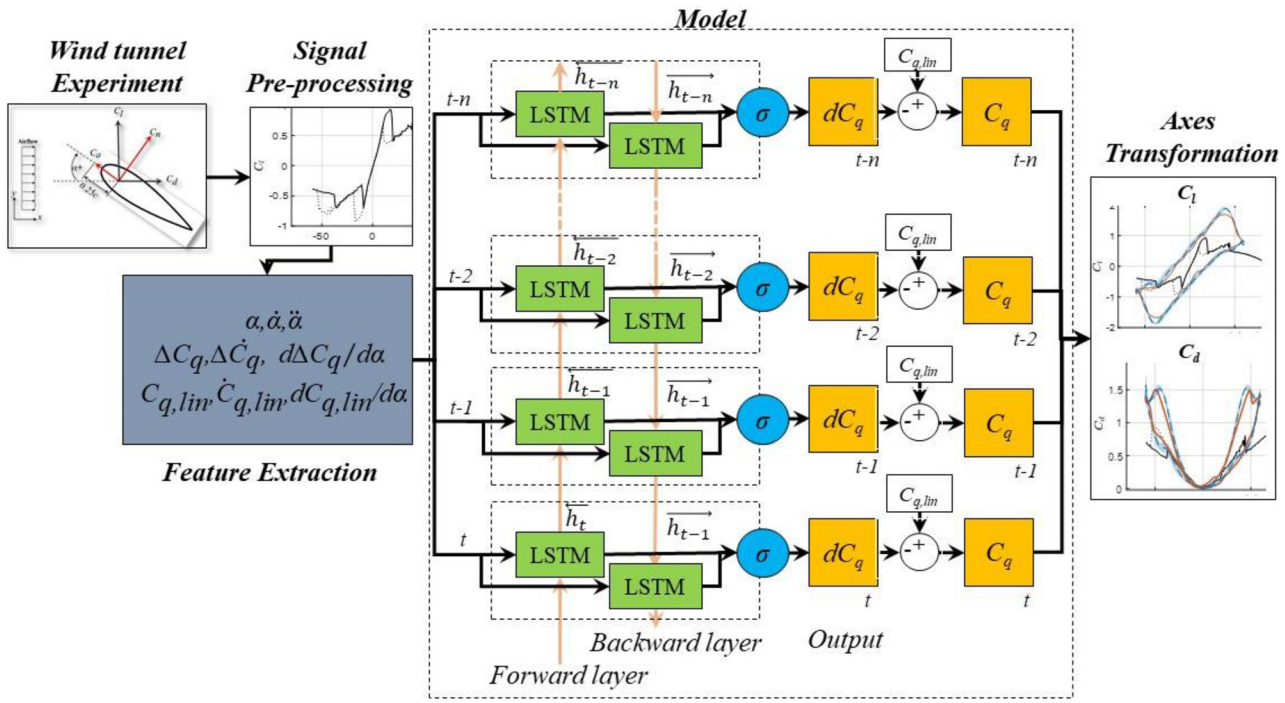


FIG. 5. Flow diagram of the proposed dynamic stall model based on the BiLSTM algorithm. The subscript q can be n or a to represent C_n or C_a, respectively.

measurements were done at $R_e = 120\,000$, lower than in, Ref. 4 to achieve load cases with a high reduced frequency without impacting the experimental setup dynamics under deep dynamic stall conditions, where load overshoots occur due to the shedding of multiple dynamic stall vortices.

In Fig. 2(a), steady C_l measurements agree with Holst *et al.*[4] in the attached flow regime below $\alpha = 18°$ in having a very similar slope but slightly lower maximum C_l at α_{ss} for the present measurements which can be attributed, inter alia, to different experimental setups, measuring the loads directly on the entire span of the airfoil model, and not using pressure transducers. When the angle of attack is greater than 150°, the steady C_l shows good agreement with prior measurements, as shown in Fig. 2(a). The steady C_d measurement shows higher drag at low α compared to the pressure measurements of Holst *et al.*,[4] as shown in Fig. 2(b), due to the inclusion of the viscous drag in the present work, as the aerodynamic forces were measured directly using force transducers. The NACA 0021 airfoil model experiences three static stalls over the entire range of angle of attack, namely, the first, second, and third or reverse flow stalls, as shown in Fig. 3, due to different mechanism of the separation point progression in these regimes as the airfoil model is pitched.[41–43] There is a clear discrepancy between measurements and literature for both steady C_l and C_d during the three stalls, particularly the strength of the second and reverse stall, as shown in Fig. 2, which depends on the configuration of the wind tunnel and experimental setup.[42,44] When steady load hysteresis is used as an input, dynamic stall models become more accurate, particularly during the pitch-down motion.[45,46] For the three hysteresis regimes indicated in Fig. 3, the angles α_1, α_2, and α_3 determine when to use the pitch-down curves. If the airfoil model was pitched

TABLE I. Wind tunnel measurement schedule for the NACA 0021 airfoil. Every data set consists of six measurements at different mean angles of attack except data set 19 used for testing has one corrupted measurement that was not used in testing. $Re = 12\,000$ for all measurements.

Purpose	Set	k	α_a^o	α_m^o
Training	1	0.0125	20	0, 15, 30, 50, 90, and 130
	2	0.075	20	0, 15, 30, 50, 90, and 130
	3	0.025	30	0, 15, 30, 50, 90, and 130
	4	0.05	40	0, 15, 30, 50, 90, and 130
	5	0.0125	60	0, 15, 30, 50, 90, and 130
	6	0.075	60	0, 15, 30, 50, 90, and 130
Validation	7	0.05	30	0, 15, 30, 50, 90, and 130
	8	0.025	40	0, 15, 30, 50, 90, and 130
Testing	9	0.0125	10	0, 15, 30, 50, 90, and 130
	10	0.025	10	0, 15, 30, 50, 90, and 130
	11	0.05	10	0, 15, 30, 50, 90, and 130
	12	0.075	10	0, 15, 30, 50, 90, and 130
	13	0.025	20	0, 15, 30, 50, 90, and 130
	14	0.05	20	0, 15, 30, 50, 90, and 130
	15	0.0125	30	0, 15, 30, 50, 90, and 130
	16	0.075	30	0, 15, 30, 50, 90, and 130
	17	0.0125	40	0, 15, 30, 50, 90, and 130
	18	0.075	40	0, 15, 30, 50, 90, and 130
	19	0.025	60	0, 15, 30, 50, and 90
	20	0.05	60	0, 15, 30, 50, 90, and 130

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down from an angle greater than these specific angles, the pitch-down load curves should be used.

An example for three unsteady load measurements at $\alpha_m = 15°$, 90°, and 130°, high pitching amplitude of $\alpha_a = 60°$, and $k = 0.05$ is shown in Fig. 4 for C_l and C_d. The unsteady C_l deviates significantly from the steady curves around the first and reverse flow stalls, as shown in Fig. 4(a), when DSVs develop to generate load fluctuations for $\alpha_m = 15°$ and 130°, respectively.[47,48] The maximum deviation of the unsteady C_d occurs at 90°, as shown in Fig. 4(b). The literature commonly defines unsteady aerodynamic loads at $k = 0.05$ as a quasi-steady loads.[4,6,31] However, under this quasi-steady condition, unsteady dynamic stall vortices might develop and cause large increases in the aerodynamic loads, as shown in Fig. 4, which emphasizes the importance of modeling the unsteady loads. For instance, the unsteady C_l deviated from the steady curve by 180% at $\alpha = 26°$ for the $\alpha_m = 15°$, while the unsteady C_d is increased by 80% at $\alpha = 98°$ for $\alpha_m = 130°$.

IV. DYNAMIC STALL MODEL

A BiLSTM algorithm was developed to reproduce the variations in the unsteady forces. The BiLSTM is an advanced variant of the LSTM algorithm with two LSTMs, as shown in Fig. 5; the first is fed by the time-history data and the second by its reverse form, with the aim of improved learning of the long-term dependencies. This should increase the prediction accuracy when the amplitude of the angle variation is large.[49] The algorithm is implemented in MATLAB using the deep learning toolbox without any auxiliary coding apart from input data handling and results processing. The BiLSTM consists of three stacked layers, each with 100 neurons; and each layer is followed by a 0.5 dropout layer to prevent overfitting as much as possible. Overfitting occurs when the model is too closely aligned with the training data that it cannot respond to the validation cases. With an initial learning rate of 0.01, the Adam optimizer was used to optimize the weight parameters of the network based on the loss function of the mean squared error. The experimental data were split into training (30%), validation (10%), and testing data (60%), Table I. This relatively small amount of training data for a regression problem, was chosen to examine the model generalization and robustness over a wide range of operating conditions. The training data were chosen to cover as much as possible of the operating conditions, particularly for the reduced frequency, to cover various degrees of flow unsteadiness. The model was trained on a workstation with an Intel Core i7–10850 CPU @ 2.7 GHz, Intel UHD graphics, and 32 GB of RAM. The main computational cost for training the proposed BiLSTM was the offline computation, which is very low in comparison to typical CFD computational cost.

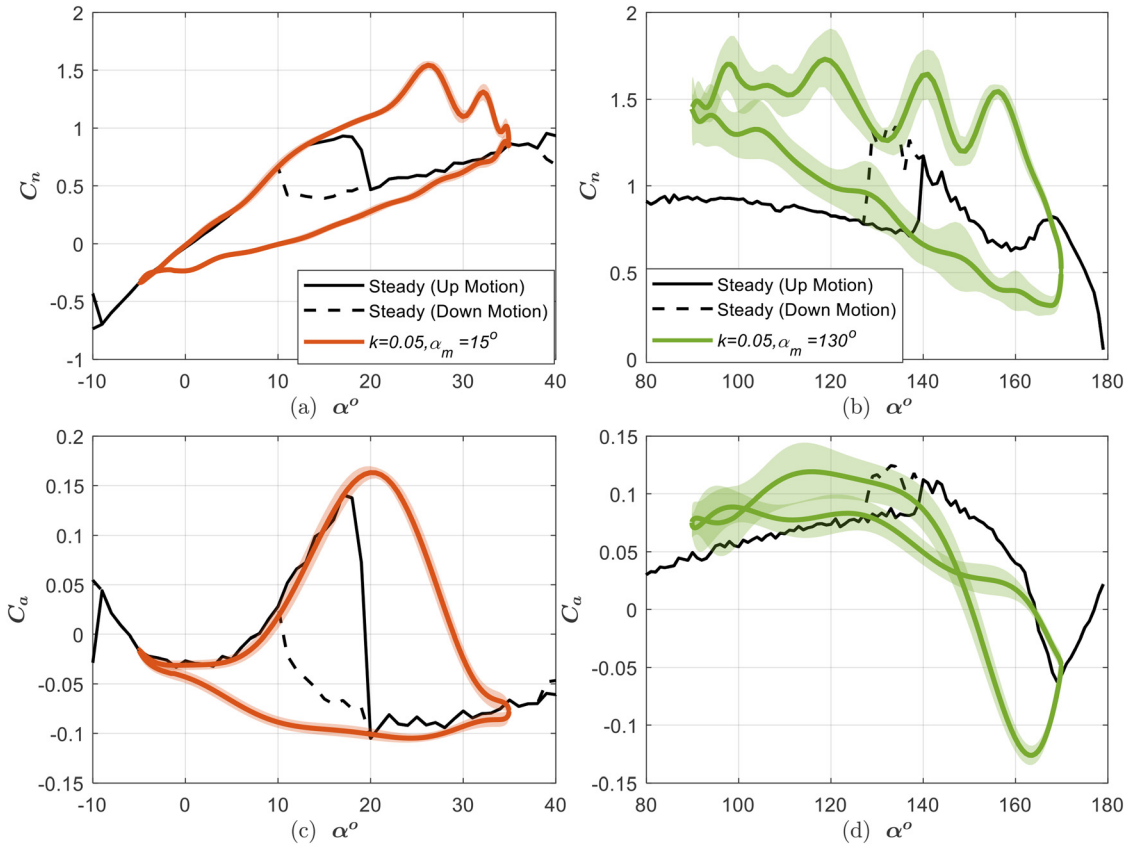


FIG. 6. Impact of vortex development and shedding on the unsteady C_n and C_a around α_{ss} and during the reverse flow: (a) C_n around α_{ss}, (b) C_n during the reverse flow and (a) C_a around α_{ss}, and (b) C_a during the reverse flow.

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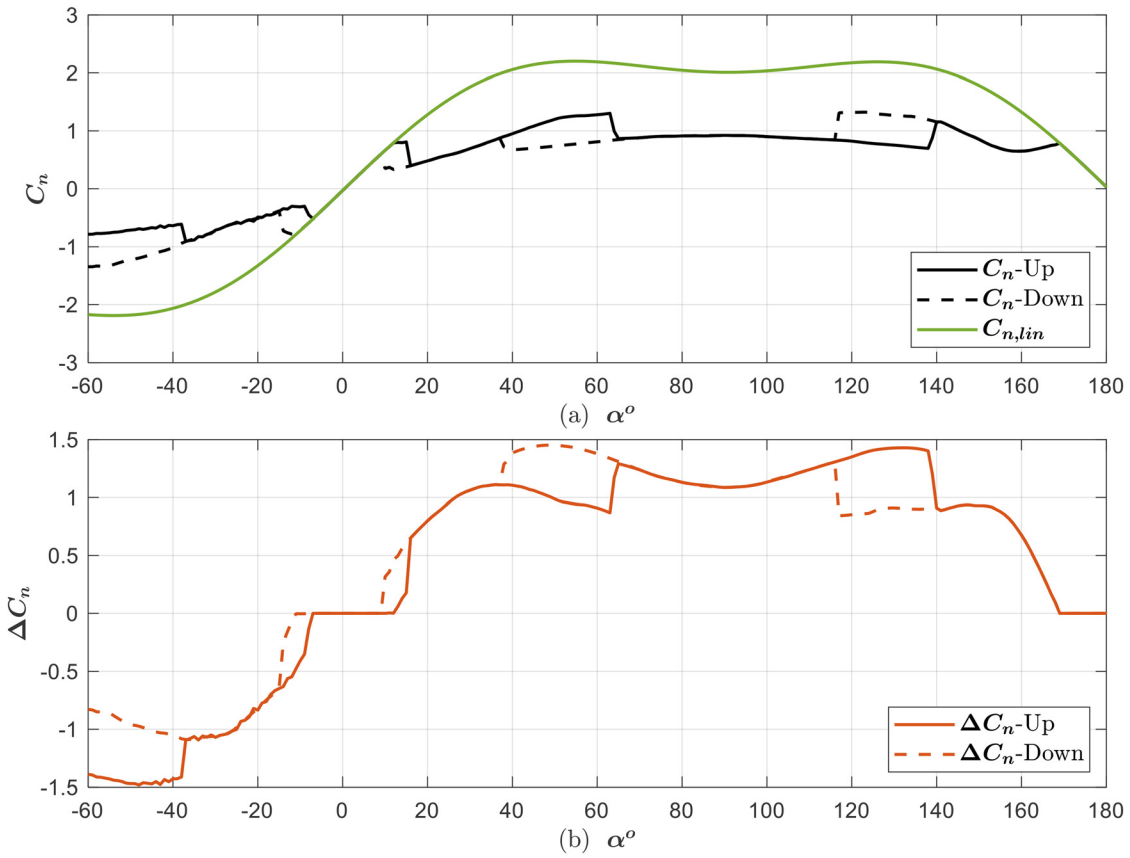


FIG. 7. Evolution of the extrapolated linear regime and the steady load loss of C_n over the entire operational range of angles of attack for the NACA 0021 at $Re = 120\,000$: (a) $C_{n,lin}$ and (b) ΔC_n.

After preprocessing the wind tunnel measurements, representative features were extracted that include three features related to the airfoil operating conditions and six features related to airfoil characteristics. The operating conditions features include α, reduced pitch rate ($\dot{\alpha}c/2V_\infty$), and reduced pitch acceleration [$\ddot{\alpha}(c/2V_\infty)^2$]. Airfoil characteristic features were extracted from the aerodynamic force coefficients in the normal- and chord-axis directions, C_n and C_a, respectively, instead of the conventional and more common C_l and C_d. Under deep dynamic stall conditions, the development and shedding of multiple DSVs can cause severe fluctuations in the unsteady C_n, C_l, and C_d, when the airfoil is pitched beyond α_{ss}.[50] Shedding of DSVs generates only a steep decrease in the unsteady C_a compared to other aerodynamic loads when the flow becomes fully separated,[2] as shown in Fig. 6. Therefore, the current model uses C_n and C_a instead of C_l and C_d to reduce the prediction effort of the BiLSTM algorithm, which can improve the overall prediction accuracy.

Airfoil characteristic features are based on the extrapolated linear regime $C_{q,lin}$ and the steady load loss ΔC_q, where the subscript q can be n or a to represent C_n and C_a, respectively. Figures 7 and 8 show the evolution of $C_{q,lin}$ and ΔC_q for C_n and C_a, respectively, for the steady pitching-up ($\dot{\alpha} > 0$) and -down ($\dot{\alpha} < 0$) motions of the NACA 0021 airfoil model. The pitching-down steady curves were included during the feature extraction stage for every load case to improve the prediction accuracy, as discussed in Sec. III. This extraction and use of airfoil characteristic features was inspired by our previous work on modeling dynamic stall forces using the ONERA dynamic stall model.[2] The inclusion of ΔC_q as an input for the BiLSTM algorithm provides implicit information about the fully separated flow regime ($\Delta C_q > 0$) and the attached flow regime ($\Delta C_q = 0$), as shown in Figs. 7(b) and 8(b). For small α, i.e., $\alpha < \alpha_{ss}$, C_l for steady flow is linear in α, and C_d is approximately constant. To extrapolate to steady flow at higher α, we use the equations for a flat plate in a deep stall[51,52] and continue to use the subscript "lin" even though the relations become nonlinear:

$$C_{d,\mathrm{lin}} = 2\sin^2\alpha + C_d(\alpha = 0), \tag{5}$$

$$C_{l,\mathrm{lin}} = 2\sin(2\alpha + \emptyset), \tag{6}$$

$$\emptyset = \tan\left(2C_l(\alpha=0)\Big/\left.\frac{dC_l}{d\alpha}\right|_{\substack{\text{attached}\\ \text{regime}}}\right). \tag{7}$$

This is followed by axes-transformation according to

$$C_{n,\mathrm{lin}} = C_{l,\mathrm{lin}}\cos\alpha + C_{d,\mathrm{lin}}\sin\alpha, \tag{8}$$

$$C_{a,\mathrm{lin}} = C_{l,\mathrm{lin}}\sin\alpha - C_{d,\mathrm{lin}}\cos\alpha. \tag{9}$$

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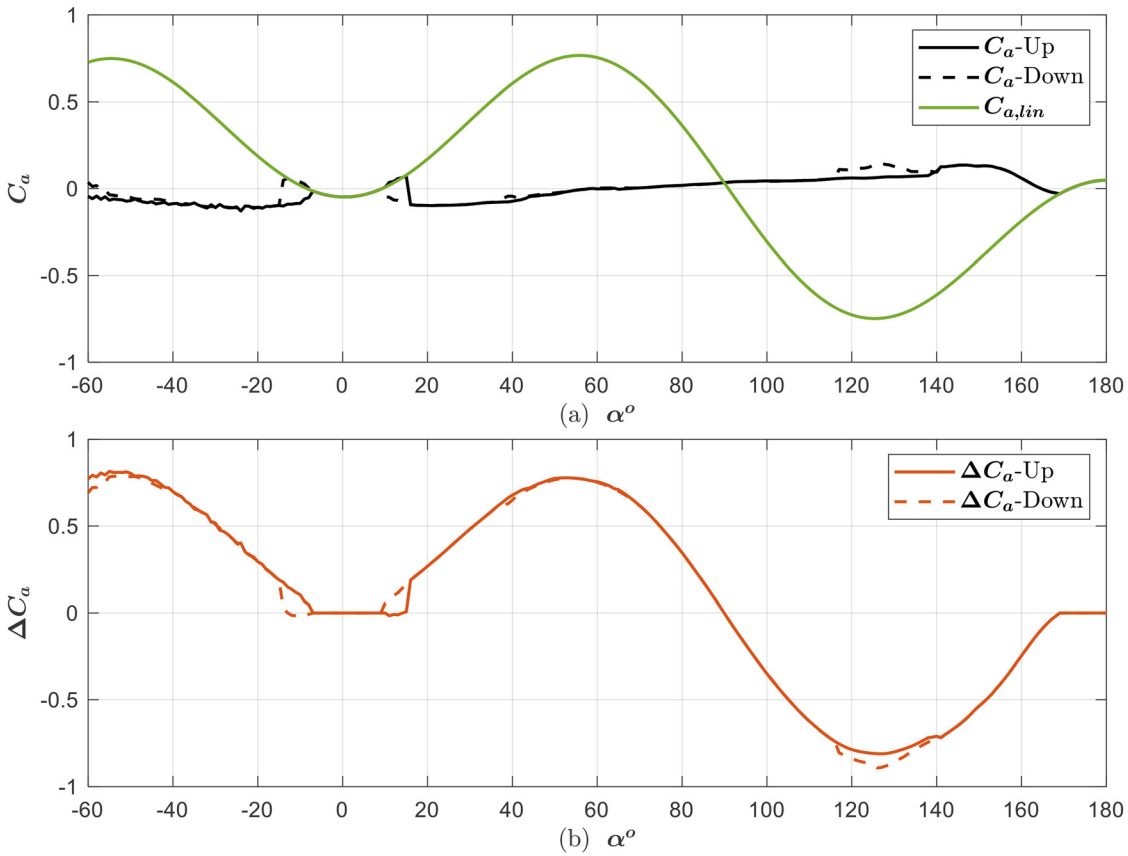


FIG. 8. Evolution of the extrapolated linear regime and the steady load loss of C_n over the entire operational range of angles of attack for the NACA 0021 at $Re = 120\,000$: (a) $C_{a,lin}$ and (b) ΔC_a.

The steady load losses ΔC_n and ΔC_a are computed according to

$$\Delta C_n = C_{n,\text{lin}} - C_n, \tag{10}$$

$$\Delta C_a = C_{a,\text{lin}} - C_a. \tag{11}$$

For every load case in Table I, $C_{n,\text{lin}}$, $C_{a,\text{lin}}$, ΔC_n, and ΔC_a were calculated, as well as their derivatives with respect to time ($^*/dt$) and α ($^*/d\alpha$).

All features are normalized to have a mean of zero and a standard deviation of unity. The BiLSTM algorithm was trained against the dynamic load loss in the normal and chord-wise directions, dC_n and dC_a, respectively, for every load case, as shown in Fig. 5. After prediction, dC_n and dC_a were subtracted from $C_{n,\text{lin}}$, $C_{a,lin}$ to compute C_n and C_a, respectively, and the axes transformation was applied to compute C_l and C_d using Eqs. (3) and (4).

V. MODELING RESULTS

The accuracy of the BiLSTM dynamic stall model is assessed in this section by comparison to the training and testing data at several

TABLE II. Summary of data plotted in Figs. 9–20. The parts of the figures show the results for α_m in the order shown in Table I.

Figure No.	Training or testing	Lift (C_l) or drag (C_d)	Set	α_a°	k
9	Training	C_l	1	20	0.012 5
10		C_d			
11		C_l	4	40	0.005
12		C_d			
13		C_l	6	60	0.075
14		C_d			
15	Testing	C_l	11	10	0.05
16		C_d			
17		C_l	17	40	0.0125
18		C_d			
19		C_l	20	60	0.05
20		C_d			

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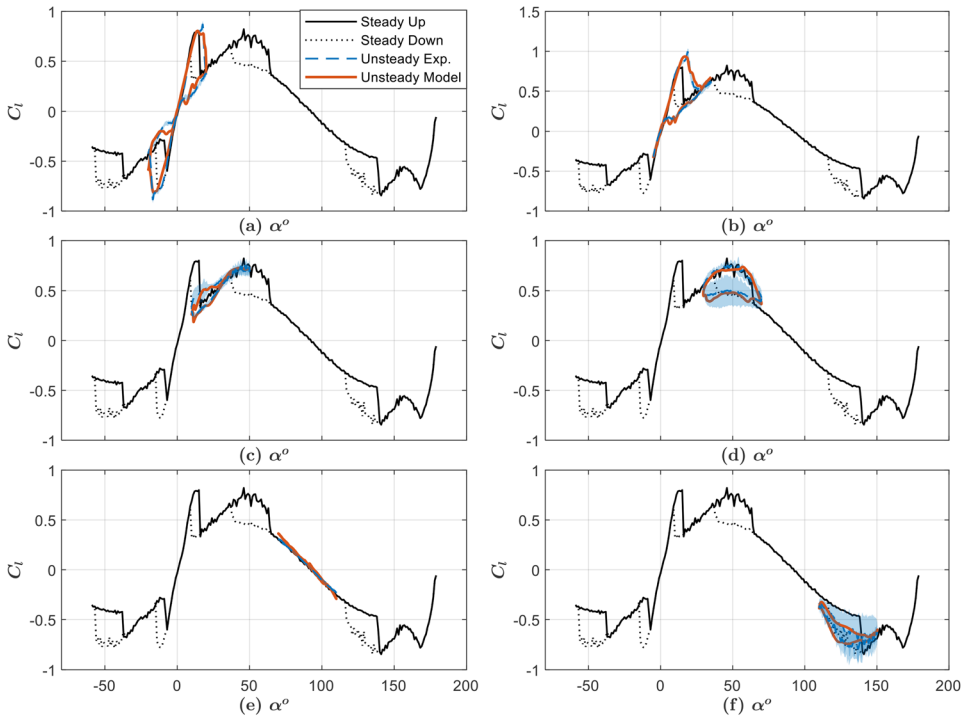


FIG. 9. Comparison between the measured and modeled C_l for training data set 1 at $\alpha = \alpha_m + 20^\circ \sin(0.0125\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

operation conditions of α_m, α_a, and k, as shown in Figs. 9–20. To help navigate through the large number of results, Table II provides the key parameters for the figures. Comparison to Table I shows we are presenting half the training data, none of the validation data, and one third of the testing data. The remaining results are provided in the supplementary material because of the large number of load cases used in the present study. Over the measured 20 oscillation cycles, the predictions of the phase averaged cycles of C_l and C_d were computed

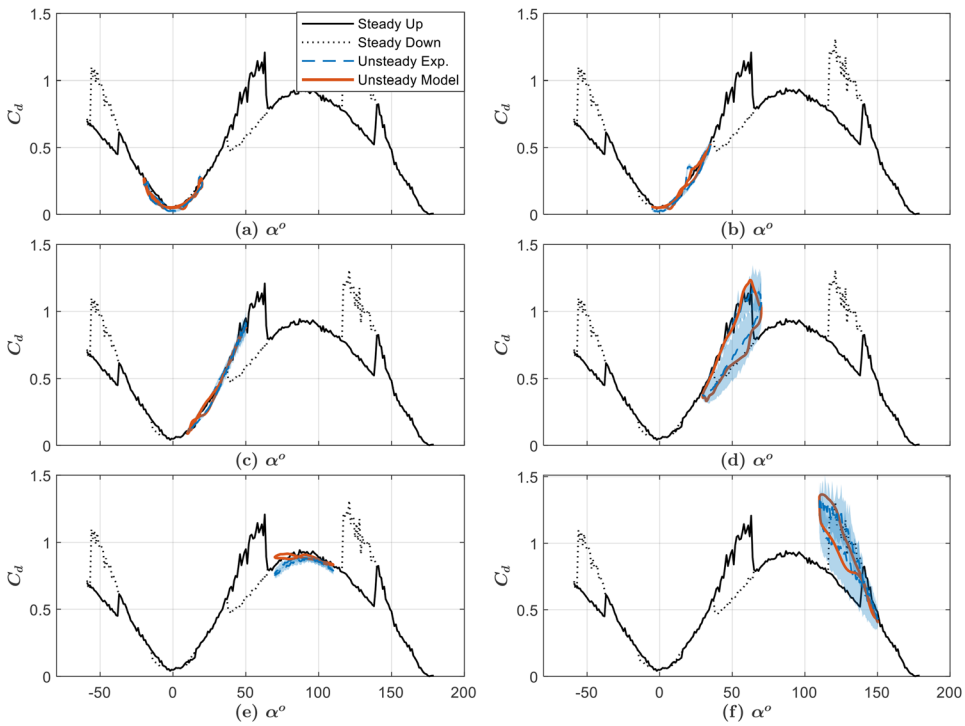


FIG. 10. Comparison between the measured and modeled C_d for training data set 1 at $\alpha = \alpha_m + 20^\circ \sin(0.0125\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

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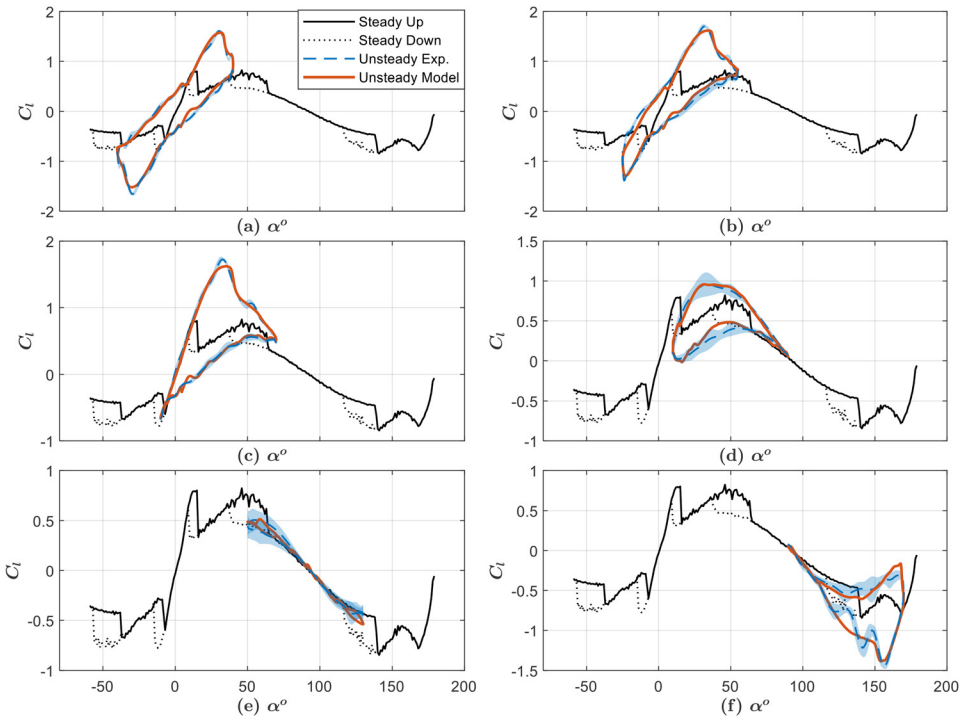


FIG. 11. Comparison between the measured and modeled C_l for training data set 4 at $\alpha = \alpha_m + 40^\circ \sin(0.05\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

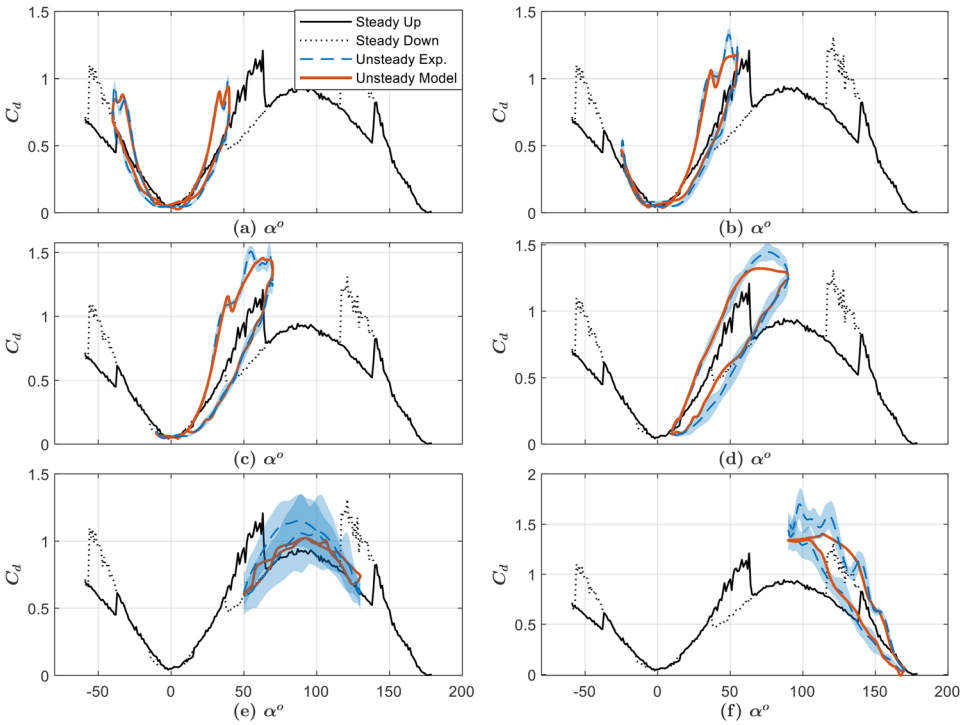


FIG. 12. Comparison between the measured and modeled C_d for training data set 4 at $\alpha = \alpha_m + 40^\circ \sin(0.05\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

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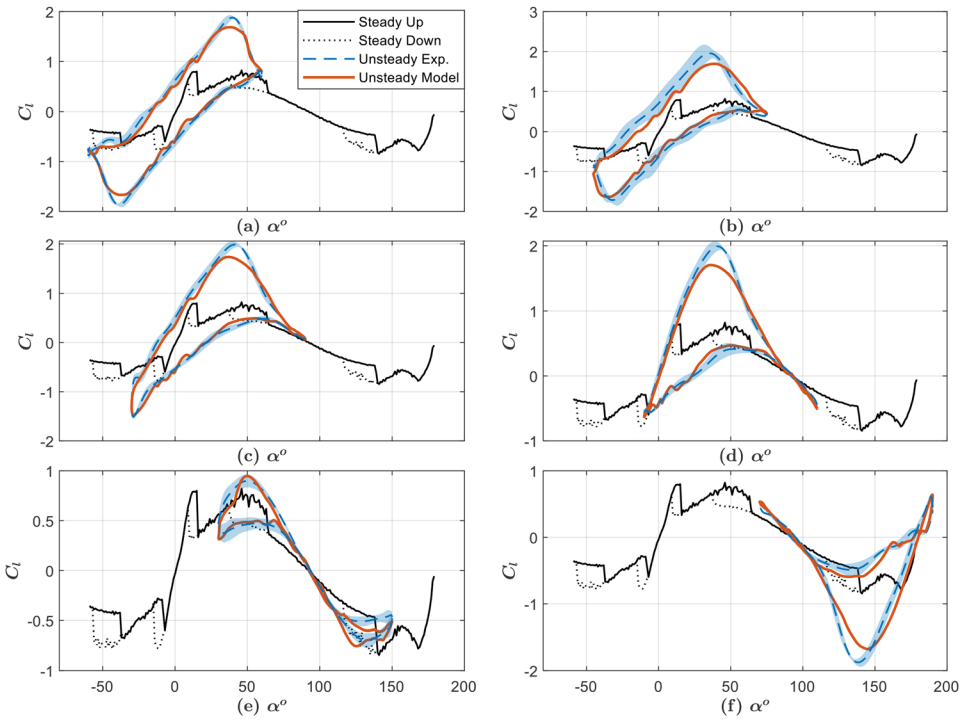


FIG. 13. Comparison between the measured and modeled C_l for training data set 6 at $\alpha = \alpha_m + 60^\circ \sin(0.075\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

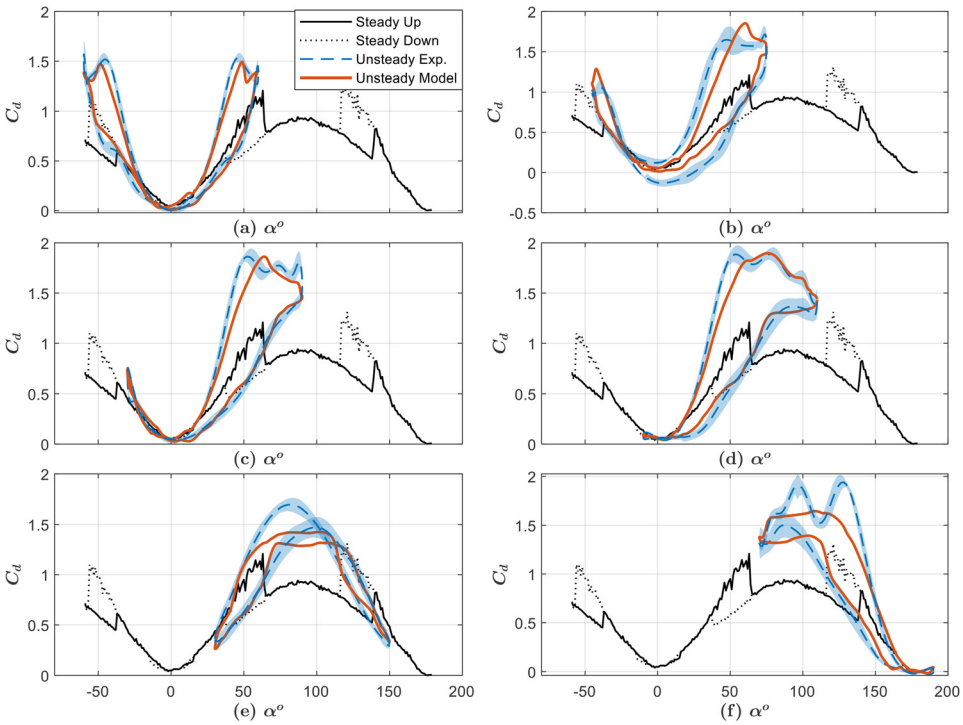


FIG. 14. A comparison between the measured and modeled C_d for training data set 6 at $\alpha = \alpha_m + 60^\circ \sin(0.075\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

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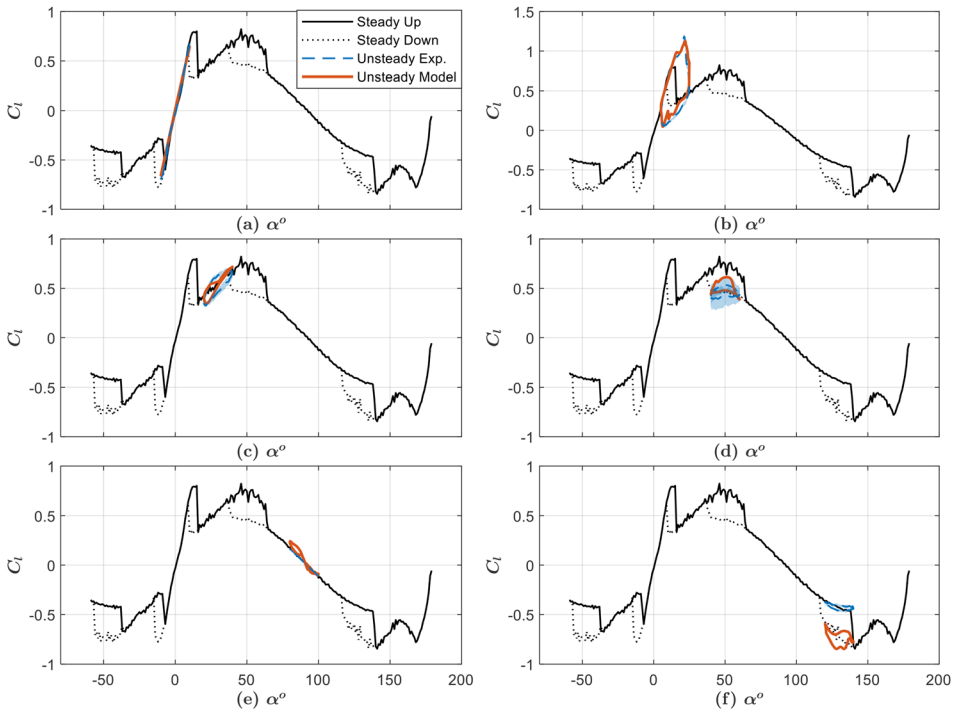


FIG. 15. A comparison between the measured and modeled C_l for testing data set 11 at $\alpha = \alpha_m + 10^\circ \sin(0.05\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

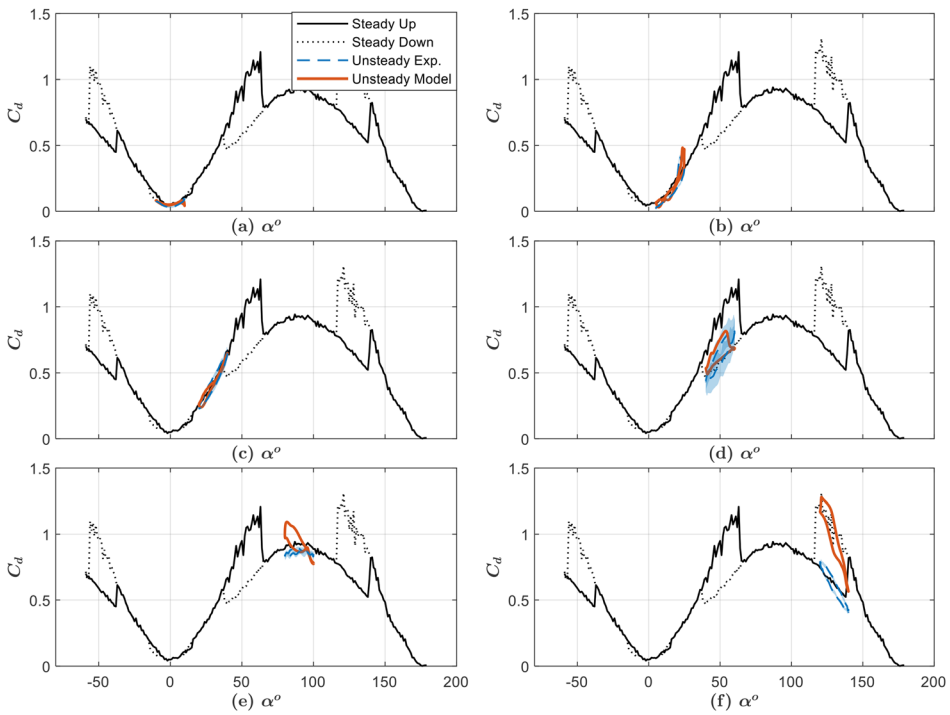


FIG. 16. A comparison between the measured and modeled C_d for testing data set 11 at $\alpha = \alpha_m + 10^\circ \sin(0.05\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

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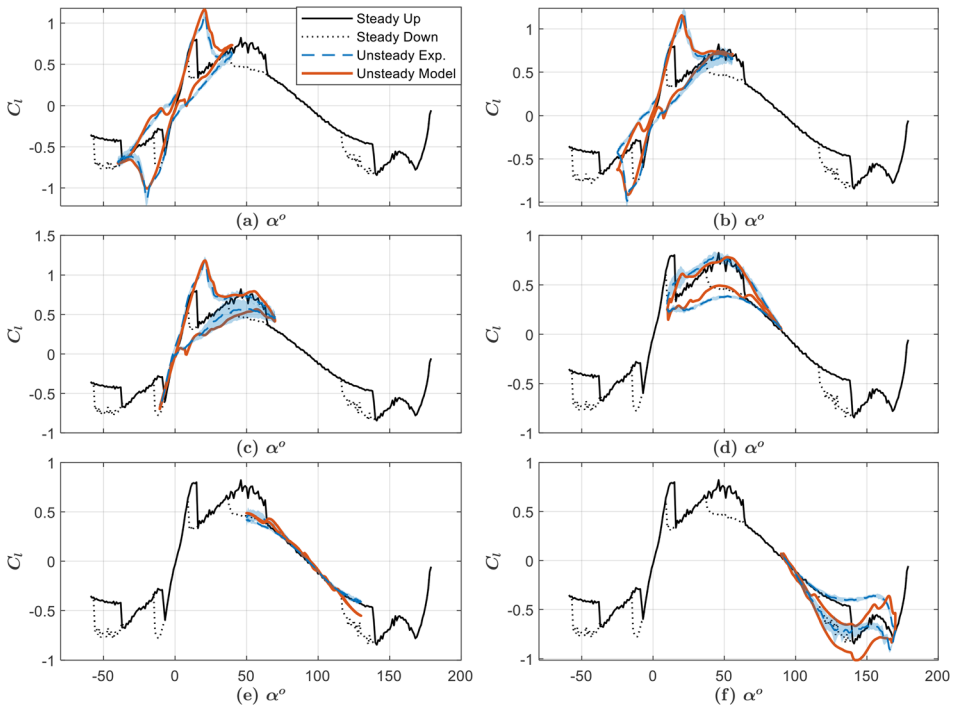


FIG. 17. A comparison between the measured and modeled C_l for testing data set 17 at $\alpha = \alpha_m + 40^\circ \sin(0.0125\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

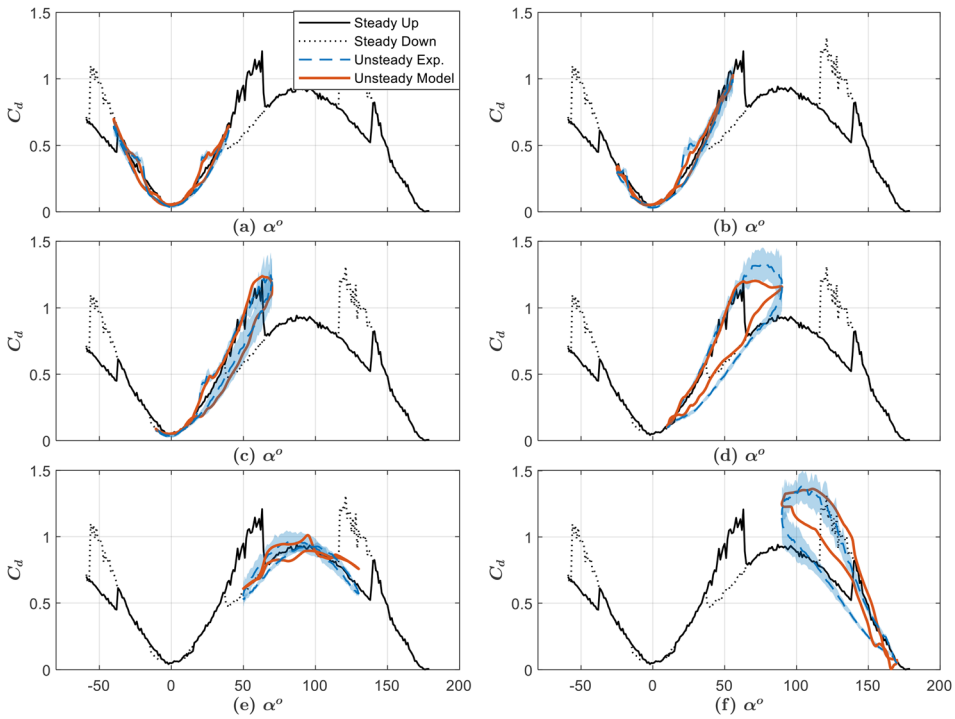


FIG. 18. A comparison between the measured and modeled C_d for testing data set 17 at $\alpha = \alpha_m + 40^\circ \sin(0.0125\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

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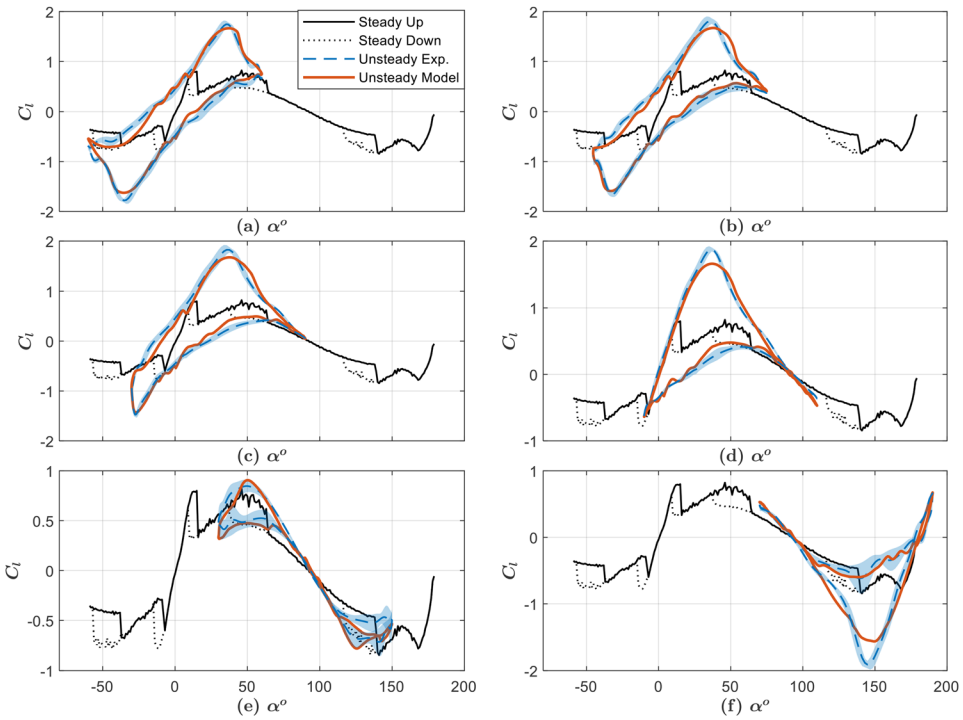


FIG. 19. A comparison between the measured and modeled C_l for testing data set 20 at $\alpha = \alpha_m + 60^\circ \sin(0.05\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

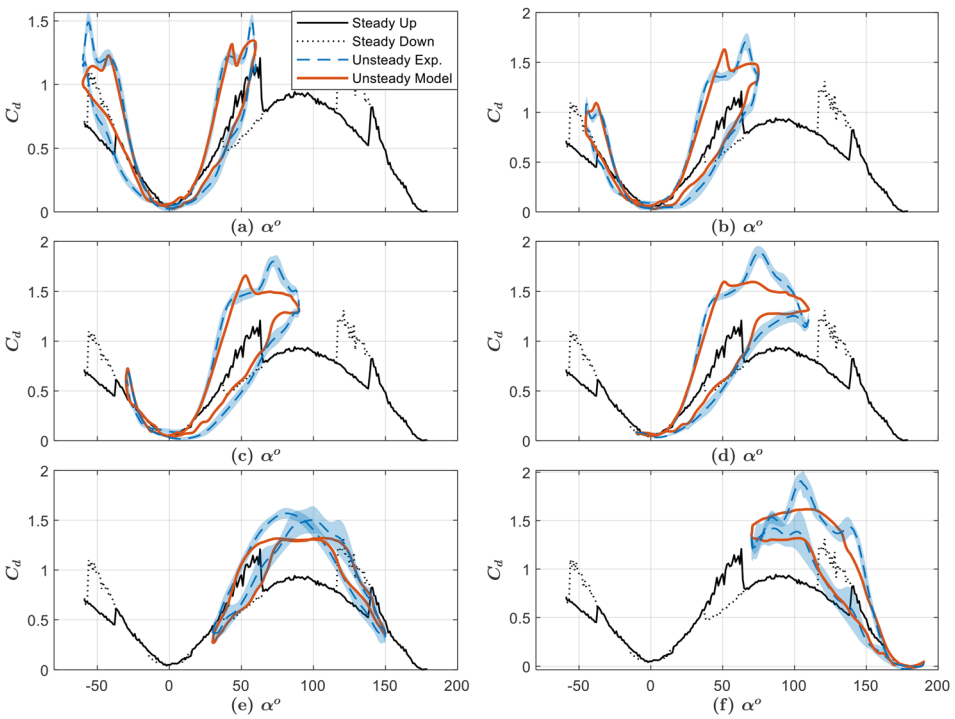


FIG. 20. A comparison between the measured and modeled C_d for testing data set 20 at $\alpha = \alpha_m + 60^\circ \sin(0.05\tau - \pi/2)$: (a) $\alpha_m = 0^\circ$, (b) $\alpha_m = 15^\circ$, (c) $\alpha_m = 30^\circ$, (d) $\alpha_m = 50^\circ$, (e) $\alpha_m = 90^\circ$, and (f) $\alpha_m = 130^\circ$.

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along with one standard deviation of the mean measurement by transforming the axes from C_n and C_a.

The training and validation results are ordered in terms of increasing k and α_a. Generally, we expect the unsteady forces to increase in complexity with an increase in both. For the lowest values in Fig. 9, the model for C_l was trained to accurately follow the measurements for all α_m. In particular, the hysteresis seen in the responses in parts (a)–(d) was well reproduced as was the response in part (e) which, like experiments confined to the linear range of α when it is close to zero, shows no hysteresis. Figure 10(a) shows that hysteresis in C_l does not necessarily carry over to drag. We note that the cyclic response associated with the second [part (d)] and third stalls [part (e)] are well captured. Increasing k and α_a in Figs. 11 and 12 increases the difference between the pitch-up and pitch-down results, which the model has been trained to capture except for the largest α_m in part (f) of both figures. The problem is associated with the high frequency oscillations in the forces that are well-captured at low α_m for the drag in Fig. 12(a), apparently cause the model to also produce them at the highest α_m. Figures 13 and 14 show the largest amplitude training results. There is significant hysteresis displayed in all parts of both figures and the model again has incorrect higher frequency oscillations at the largest α_m. This occurs in the range of α where C_l is smaller than C_d, so the problem is not the relative magnitude of the two forces. The overshoot in C_l as α increases is associated with the shedding of the primary DSV, whose significance clearly increases with increasing α_m.

We now consider the validation data presented in the same order, but, of course with different values of k and α_m. Figures 15(a) and 16(a) demonstrate that the model reproduces the lack of hysteresis when α is confined to the linear range that was noted above. The model predicts the cyclic lift well for all but the highest α, and similarly for the drag for all but the two highest α. For $\alpha_m = 130°$, both the modeled lift and drag follow the steady trajectory for decreasing α whereas the measurements follow the increasing α trend. It is possible that the model could not be improved until we have a better understanding of the differences between the steady forces for increasing and decreasing α. The model is more accurate at high α when α_a increases, Figs. 17 and 18, suggesting that the errors in the model just referred to, are due to restriction of the cycle to α-range where the steady hysteresis is important. Generally, the model is accurate at the highest α_m, Figs. 19 and 20, except that the high frequency oscillations noted above have reappeared in the drag in all parts of Fig. 20. It is also noted that the unsteadiness in the response of the lift and drag increases with α_m which may complicate the modeling of the response.

Given that the basic structure of the model was not altered during the training over a wide range of k and α_m, its general ability to match the measured lift and drag for the sinusoidally pitched NACA 0021 section has been demonstrated to be good. The lift has been modeled more accurately than the drag, where the high-frequency oscillations needed at low α are amplified at high α and reduce the model accuracy.

VI. CONCLUSION

The key contribution of the present research is to explore the potential of a deep learning framework to predict the nonlinear unsteady forces on oscillating airfoils over the entire range of operation for wind turbine applications. Training, validation, and testing data were obtained from wind tunnel measurements for an oscillating NACA 0021 airfoil model at a Reynolds number of 120 000 with oscillation amplitude up to 60° and reduced frequency up to 0.075. By training the model using 30% of the unsteady force measurements, the deep learning BiLSTM algorithm is capable of reproducing variations in unsteady forces at angles of attack ranging from −60° to 190°. A novel set of features, inspired by empirical models, was constructed to represent the operating conditions and airfoil characteristics. Accurate predictions were achieved in all regimes except when reproducing the load overshoots generated by the shedding of multiple dynamic stall vortices.

The model can be easily incorporated in wind turbine design codes; however, further research should be done to expand and corroborate these preliminary findings by reproducing the impact of the shedding of multiple dynamic stall vortices on the unsteady loads around the static stall angle and in the reverse flow regime. The future research should also include an investigation of the accuracy and generality of the model to reproduce unsteady loading for other airfoil shapes at high angles of attack.

SUPPLEMENTARY MATERIAL

Remaining load cases listed in Table I and the associated BiLSTM predictions are provided in the supplementary material.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Conceptualization, A.M., and D.W.; methodology, A.M; software, A.M.; validation, A.M. and D.W.; formal analysis, A.M.; investigation, A.M.; resources, D.W.; data curation, A.M.; writing—original draft preparation, A.M.; writing—review and editing, A.M. and D.W.; visualization, A.M.; supervision, D.W.; project administration, A.M. and D.W.; funding acquisition, D.W. All authors have read and agreed to the published version of the manuscript.

Ayman Salah Ashry Mohamed: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Project administration (equal); Software (lead); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **David Wood:** Conceptualization (equal); Funding acquisition (lead); Project administration (equal); Resources (lead); Supervision (lead); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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